

ArtifactBench: Lineage-Aware Evaluation of AI-Generated Music Detectors under Distribution Shift

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Abstract

AI-generated music detectors are commonly compared using aggregate scores on benchmarks whose training overlap, generator lineage, source provenance, and audio-transformation history are only partially observable. This paper introduces ArtifactBench, a lineage-aware evaluation suite for measuring detector behavior across generator families and versions, real-music domains, collection-cohort shift, and inference coverage. The benchmark groups source recordings and their derived variants by content identity, separates calibration from final testing, records inference failures independently from classification errors, and reports source-level performance with uncertainty in addition to aggregate metrics. We evaluate multiple publicly available detectors under a version-pinned common protocol and examine how leakage control, cohort availability, threshold policy, and model-specific missingness alter measured performance and model ranking. On the 562-track common-success test intersection, ArtifactNet obtains 0.982 AUROC and 0.918 balanced accuracy, compared with 0.761/0.776 for the public Deezer detector; SpecTTRa and CLAM fall below 0.30 AUROC under this shifted cohort. These results also expose substantial generator- and real-domain shifts that aggregate scores alone conceal.

1 Introduction

AI-generated music detection is increasingly evaluated under distribution shift: new generators, changing model versions, post-processing, and real music drawn from domains unlike the detector’s training data. Recent datasets have made the task measurable at useful scale [11, 3, 10], while detector studies have repeatedly shown that high in-domain accuracy can coexist with weak robustness to unseen generators or audio transformations [2, 1]. A less visible source of variation is the evaluation cohort itself. A benchmark name may refer to several manifests, only a subset of referenced audio may still be available, and a model’s public training split may overlap the nominal test set. If each adapter silently skips a different set of files, the resulting aggregate scores do not compare the same population.

This paper studies AI-music detection as an evaluation-governance problem. We introduce ArtifactBench v2, a lineage-aware protocol that binds every public track identifier to a content digest, source family, generator version when known, and a lineage identifier. It separates cohort construction from model execution, freezes calibration, validation, and test partitions before final analysis, and records decoding and inference failures separately from wrong predictions. A common operating point is selected on calibration data alone; threshold-free and model-native results remain visible so that conclusions do not depend on one threshold convention.

Our contributions are:

- a forensic audit of three incompatible ArtifactBench manifest lineages and the audio bytes that can actually be reconstructed;
- a content- and lineage-controlled 828-track primary cohort spanning 13 synthetic source strata and two real-music domains, split into calibration, validation, and sealed test partitions without lineage conflicts;
- a version-pinned runner that applies one deterministic cohort to four public detectors, preserves raw track-level scores, and treats inference coverage as a reported result; and

- an analysis of source and generator-version shift, threshold policy, upstream split exposure, and their effect on measured performance and model ranking.

ArtifactBench v2 is not proposed as the largest training corpus, nor as a claim that every track is unknown to every representation model. Its goal is narrower: to make the population, provenance, operating point, and failure denominator of each comparison auditable.

2 Related Work

2.1 AI-Generated Music Detection

SONICS introduced a large full-song corpus and the SpecTtTra family of spectrogram transformers [11]. Deezer demonstrated that a detector trained from artificial reconstructions can obtain very high held-out accuracy while emphasizing manipulation and unseen-generator risks [2]; later work connected detection evidence to Fourier structure [1]. MoM and CLAM combine music and speech-oriented representations under contrastive training [3]. ArtifactNet instead operates on estimated codec residuals and harmonic/percussive structure [9]. MusicDET frames the task as real-only one-class density estimation [7]. These methods differ not only in architecture but also in crop duration, aggregation, training corpora, and score calibration; comparing published headline values therefore conflates model and protocol.

2.2 Audio Deepfake Benchmarks

Echoes controls semantic alignment between bona-fide references and generated tracks to reduce shortcut learning [10]. HAIM expands the label space beyond binary origin to stages of human–AI production interaction [5]. BAMB shows a further domain gap in real television broadcast audio [8]. These datasets expose complementary axes—semantic alignment, production provenance, and deployment channel. Our focus is the measurement layer shared across such axes: track identity, train–test exposure, missingness, threshold selection, and paired comparison.

2.3 Dataset Lineage, Leakage, and Distribution Shift

Content leakage is stronger than a duplicated filename and weaker than an exact byte match alone. Re-encoding, trimming, or alternate containers may preserve the same recording while changing its digest. ArtifactBench therefore uses both a cryptographic content identity and a lineage group. The split unit is the lineage, not the file. We also distinguish (i) exact content overlap, (ii) membership in a published upstream train or validation partition, and (iii) unresolved exposure where the public training procedure does not provide an identity map. These states are reported rather than collapsed into a binary “unseen” label.

3 ArtifactBench Design

3.1 Scope and Intended Use

The primary task is binary track-level detection of fully generated music versus human-produced music. The benchmark is intended for comparative research, failure analysis, and reproducibility testing. It does not establish authorship, copyright status, intent, or whether a partially AI-assisted production should be labeled synthetic. Segment localization, hybrid-production tracking, and commercial adjudication are outside the primary task.

3.2 Source Provenance and Licensing

The primary cohort contains 605 synthetic and 223 real tracks. The synthetic side includes 310 tracks from the AIME release [6], 180 Suno CDN or supplementary tracks, and 115 Udio CDN or supplementary tracks. The real side contains 150 tracks recovered from the official FMA large archive and 73 hard negatives referenced by stable source identifiers. FMA metadata was matched for all 150 selected IDs [4].

Table 1: Frozen primary cohort before protocol partitioning.

Source group	Strata	Tracks
AIME generators	9	310
Suno supplementary/CDN	2	180
Udio supplementary/CDN	2	115
FMA real hard negatives	1	150
Web real hard negatives	1	73
Total	15	828

Because source licenses differ and redistribution rights for platform-collected audio are not uniformly established, the v2 public package is metadata-first: it contains path-free identifiers, digests, source provenance, split assignments, and reconstruction instructions, but does not introduce a new audio bundle. The private runtime manifest binds those public identities to locally verified bytes and is not released.

3.3 Content Identity and Lineage Groups

Each row contains a stable `track_id`, binary label, source stratum, generator family and version when available, SHA-256 of the evaluated bytes, and a `lineage_id`. Exact byte matches against recovered ArtifactNet training audio were removed. Two real-audio variants found by acoustic fingerprinting were also removed. A lineage may contain several containers or transformations, but no lineage may cross protocol partitions.

3.4 Generator, Domain, Time, and Codec Axes

Table 1 summarizes the primary population. AIME sources cover open and hosted generation systems available during its collection period; the supplementary Suno and Udio strata add later or independently collected outputs. FMA provides artist-licensed 30-second real music, whereas the second real stratum emphasizes web-sourced hard negatives. These labels are retained in every raw prediction to prevent aggregate performance from hiding a source collapse.

3.5 Calibration, Validation, and Sealed Test Partitions

We assign lineages deterministically within source strata using a published salt and target proportions of 20/10/70. This yields 166 calibration tracks (121 AI, 45 real), 83 validation tracks (61 AI, 22 real), and 579 sealed test tracks (423 AI, 156 real). The split was written to the manifest before the final four-model analysis. Calibration selects operating thresholds; validation is diagnostic only and cannot trigger retuning; test is used once for the reported comparison.

4 Evaluation Protocol

4.1 Version-Pinned Model Adapters

We evaluate ArtifactNet v9.4, SpecTTRa- α -120s, the public Deezer ISMIR 2025 logistic detector, and CLAM. Repository, model, and feature-encoder revisions are pinned rather than fetched from mutable default branches. ONNX models are materialized with adjacent external-data files and their revisions are recorded in each run manifest. CLAM additionally records the checkpoint SHA-256 and the pinned MERT and Wav2Vec2 encoder revisions.

4.2 Common and Model-Native Preprocessing

All adapters receive the same decoded mono waveform and selected track IDs in the same deterministic order. Resampling occurs inside each adapter to its native sampling rate. Fixed-duration baselines use a center crop, not a random crop. SpecTTRa uses its 120-second input policy; CLAM uses a deterministic 90-second crop.

ArtifactNet evaluates seven evenly spaced four-second chunks and uses the median of finite chunk scores when at least four of seven chunks are valid. We retain a strict sensitivity result in which any non-finite chunk causes a track failure and publish the chunk-level failure log. Four is the smallest strict majority; this source-agnostic reliability rule was fixed before the sealed-split analysis and was not optimized against labels or performance. No model-specific replacement track is introduced after a decode or inference failure.

4.3 Threshold Policies

The primary operating threshold for each model is chosen using calibration rows only. Among observed score thresholds with false-positive rate (FPR) no greater than 5%, we maximize true-positive rate (TPR); ties select the lower threshold. The threshold is then frozen for validation and test. We separately report the native 0.5 threshold and threshold-free area under the ROC curve (AUROC) and average precision (AUPRC). This separation prevents a favorable calibration choice from being presented as representation quality.

4.4 Inference Failures and Missing Data

Resolution, decode, model inference, non-finite output, and out-of-range probability failures are distinct events. They are not converted into a class prediction and are not silently dropped from the attempted denominator. Main tables show both successfully scored tracks and failures. Pairwise comparisons use the common successfully scored test-ID intersection; coverage itself remains a model outcome.

4.5 Metrics and Statistical Uncertainty

For each partition and source we report TPR or FPR, and for mixed-label partitions AUROC, AUPRC, F1, balanced accuracy, and the confusion matrix. Ninety-five percent intervals are computed with 2,000 label-stratified lineage bootstrap replicates. Resampling the lineage rather than the file prevents multiple encodings of one recording from being treated as independent evidence.

5 Experiments

5.1 Models and Reproducibility Environment

All final inference runs are executed on one Apple Mac Studio (M1 Max, 32 GB) from a frozen runtime copy whose 828 content hashes were reverified. PyTorch models use MPS where supported; the ArtifactNet ONNX model uses CPU execution. Each result directory contains the input-manifest digest, selected track IDs, seed, package versions, platform string, model provenance, per-track scores, and structured failures.

5.2 Leakage and Cohort Audit

The public v1.0.1 manifest contains 6,200 rows and a 2,280-row test partition; the public v1.1 branch contains a 2,224-row purged test set; and a local post-purge manifest contains 6,014 actual rows despite metadata describing 6,183. An earlier eight-model comparison scored only 2,104 tracks because 120 referenced FMA files were HTML responses rather than audio. We recovered all 150 selected FMA tracks from the official archive and verified their identities.

The lineage-clean intermediate cohort contains 2,236 rows. Against public SONICS lineage, 241 rows map to train or validation, 454 to test only, and 1,541 are external. Against public MoM lineage, 200 map to its training partition, 197 to its test partition, 1,523 are external, and 316 real rows remain unresolved because the released CLAM training code applies a seeded random split after local embedding-availability filtering without publishing the realized identity map. The conservative 828-track primary cohort excludes all native SONICS- and MoM-derived source families. We still avoid describing it as universally training-unseen.

Table 2: Paired test performance. Metric columns use the 562-track common-success intersection; coverage uses all 579 attempted test tracks. Thresholds were selected on calibration data only.

Model	scored	τ	AUROC	AUPRC	F1	BACC	TPR	FPR	coverage
ArtifactNet v9.4	562/579	0.9909	0.982	0.994	0.923	0.918	0.865	0.029	0.971
SpecTtTra- α	579/579	0.9695	0.299	0.688	0.159	0.526	0.087	0.036	1.000
Deezer ISMIR	579/579	0.3165	0.761	0.925	0.758	0.776	0.624	0.072	1.000
CLAM	579/579	1.0000	0.284	0.647	0.000	0.500	0.000	0.000	1.000

Table 3: Ninety-five percent lineage-bootstrap intervals (2,000 replicates) on the paired test intersection.

Model	AUROC	AUPRC	TPR	FPR
ArtifactNet v9.4	0.971–0.991	0.989–0.997	0.832–0.898	0.007–0.058
SpecTtTra- α	0.245–0.352	0.662–0.714	0.061–0.116	0.007–0.072
Deezer ISMIR	0.723–0.798	0.912–0.938	0.579–0.671	0.029–0.115
CLAM	0.236–0.333	0.623–0.675	0.000–0.000	0.000–0.000

6 Results

All values in this section are generated from the frozen per-track JSON bundle. No number is transcribed from the earlier ArtifactNet paper or the 2,104-track exploratory comparison.

6.1 Paired Aggregate Performance

Table 2 reports classification metrics on the 562 test tracks successfully scored by all four models. The scored and coverage columns retain the full 579-track attempted denominator. ArtifactNet leads the paired comparison with AUROC 0.982, AUPRC 0.994, and balanced accuracy 0.918. The public Deezer detector is second at 0.761 AUROC and 0.776 balanced accuracy. SpecTtTra and CLAM obtain AUROC 0.299 and 0.284, respectively, showing that their score orderings do not transfer to this cohort. For CLAM, satisfying the calibration FPR constraint requires the next representable threshold above its maximum score of 1.0; the resulting operating point predicts no positives. We therefore show its threshold-free and native-threshold behavior rather than interpreting the calibrated F1 in isolation.

The lineage-bootstrap intervals in Table 3 preserve the same ordering. ArtifactNet’s AUROC interval is 0.971–0.991, whereas Deezer’s is 0.723–0.798. The non-overlap is descriptive evidence for this frozen cohort, not a universal claim over future generators.

6.2 Generator-Version and Real-Domain Shift

Table 4 separates source cohorts without treating collection membership as a precise per-track timestamp. ArtifactNet retains TPR 0.926 on AIME and 0.897 on supplementary Suno, but falls to 0.650 on supplementary Udio. Deezer shows the opposite pattern: TPR is 0.332 on AIME but rises to 0.968 on Suno and 0.875 on Udio. Its real-music FPR changes from 0.020 on FMA to 0.211 on the web hard-negative cohort. Thus neither the generator side nor the real side is represented by a single stable domain.

Figure 1 gives the corresponding source-stratum view. It makes clear that a favorable group mean can coexist with a generator-version collapse; the labels denote collection cohorts and known generator versions, not exact generation dates.

Table 4: Cohort-group rates on the common-success test intersection at each model’s calibration-selected threshold.

Model	AIME TPR	Suno TPR	Udio TPR	FMA FPR	Web FPR
ArtifactNet v9.4	0.926	0.897	0.650	0.040	0.000
SpecTtTra- α	0.097	0.103	0.037	0.050	0.000
Deezer ISMIR	0.332	0.968	0.875	0.020	0.211
CLAM	0.000	0.000	0.000	0.000	0.000

Table 5: Coverage and failure-as-error classification bound on all 579 test attempts.

Model	scored	failures	coverage	F1	BAcc	TPR	FPR
ArtifactNet v9.4	562/579	17	0.971	0.904	0.865	0.865	0.135
SpecTtTra- α	579/579	0	1.000	0.159	0.528	0.087	0.032
Deezer ISMIR	579/579	0	1.000	0.756	0.777	0.624	0.071
CLAM	579/579	0	1.000	0.000	0.500	0.000	0.000

6.3 Coverage and Numerical Reliability

The three comparison baselines score all 579 test tracks. ArtifactNet scores 562 (97.1% coverage) under the declared finite-chunk rule, leaving 17 test failures. Across all 828 tracks, 165 contain at least one non-finite chunk; 141 retain at least four of seven finite chunks and are recovered by the predeclared majority rule, leaving 24 unresolved tracks. Non-finite chunks are not confined to silence: their measured RMS ranges up to -4.69 dBFS. This points to a numerical model-path failure rather than a simple silent-input condition.

Table 5 applies an intentionally pessimistic classification bound in which every missing synthetic track is a false negative and every missing real track is a false positive. ArtifactNet’s F1 then falls from 0.923 on common successful tracks to 0.904, and its balanced accuracy to 0.865. This bound does not fabricate probabilities, so AUROC and AUPRC remain undefined for failed rows.

Under the stricter sensitivity policy that rejects a track after any non-finite chunk, ArtifactNet scores only 467 test tracks. On those tracks it obtains AUROC 0.985 and balanced accuracy 0.932, but failure-as-error balanced accuracy drops to 0.710 because 112 test attempts are unresolved. The finite-chunk rule therefore improves coverage materially without creating a better score for any failed chunk; both policies and the chunk-level log are released.

6.4 Ranking Sensitivity

ArtifactNet and Deezer rank first and second under calibrated balanced accuracy, native-threshold balanced accuracy, and AUROC (Table 6). The lower two positions change: CLAM ranks above SpecTtTra at the native 0.5 threshold, while SpecTtTra ranks above CLAM after calibration and by AUROC. More importantly, native 0.5 behavior is operationally poor for both: SpecTtTra has balanced accuracy 0.355, and CLAM 0.473 with TPR 0.889 but FPR 0.942. A single headline F1 would conceal that CLAM’s native score is driven by predicting most tracks as synthetic.

6.5 Paired Codec Robustness

Codec-pair experiments are secondary because the frozen primary run contains no derived codec variants. A later codec analysis must use identical excerpts and retain the parent lineage; it is not mixed into the primary 579-track test.

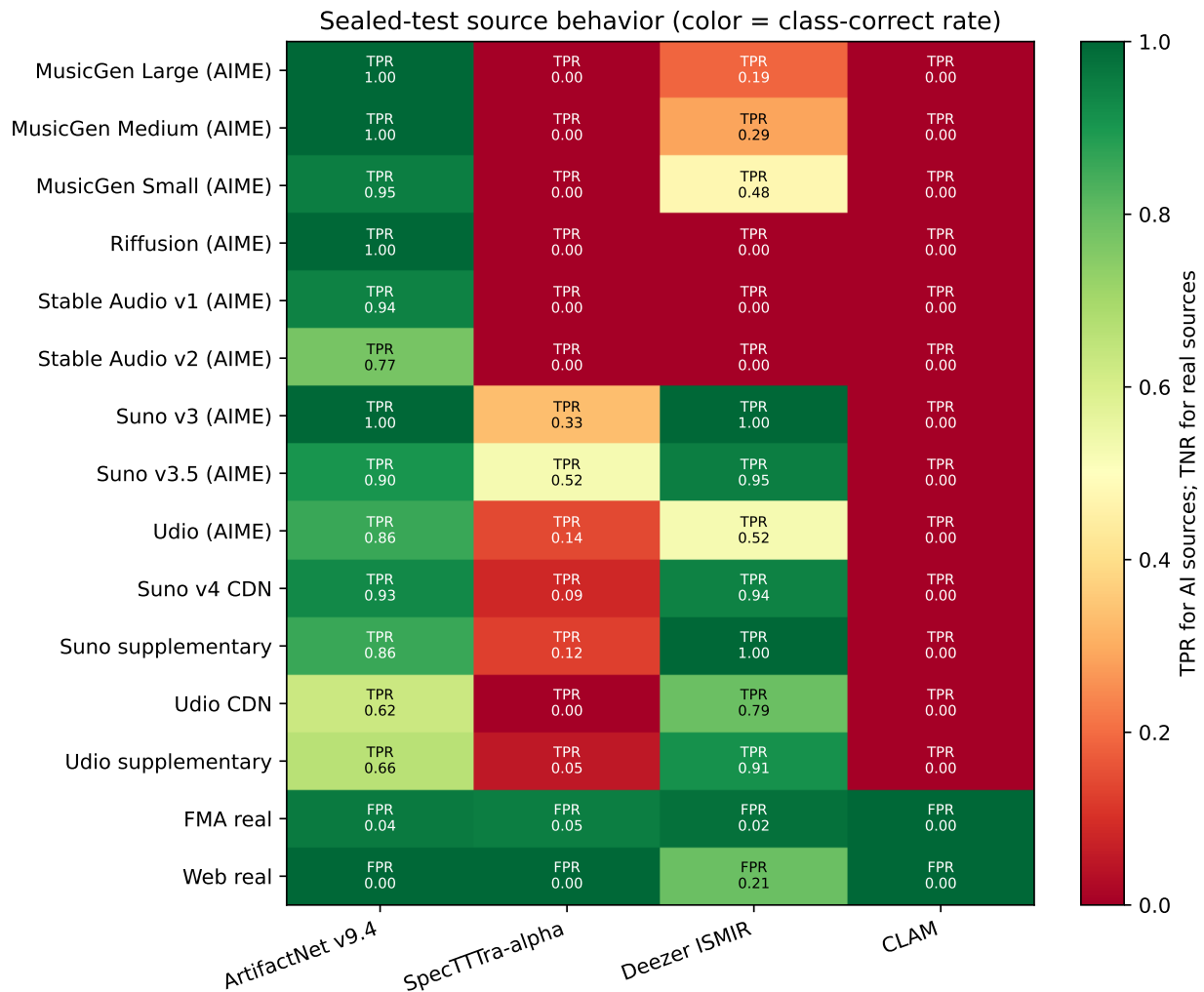


Figure 1: Per-source TPR for synthetic strata and FPR for real strata on the paired test intersection. Values use calibration-selected thresholds.

Table 6: Ranking sensitivity on the paired test intersection.

Model	calibrated BAcc	native-0.5 BAcc	AUROC
ArtifactNet v9.4	0.918	0.934	0.982
SpecTTRa- α	0.526	0.355	0.299
Deezer ISMIR	0.776	0.767	0.761
CLAM	0.500	0.473	0.284

7 Discussion

7.1 What Aggregate F1 Hides

An aggregate F1 score combines class prevalence, one operating threshold, and a particular mixture of generator and real domains. A detector can therefore appear strong while failing one real-music stratum, or weak when its ranking scores are useful but its default threshold is miscalibrated. Source-level rates and threshold-free metrics are necessary to distinguish these cases.

7.2 Benchmark Construction Bias

Benchmark curation can inadvertently reward a detector through codec, duration, sampling-rate, or source shortcuts. Semantic alignment addresses one part of this problem [10]; content lineage and split exposure address another. Neither guarantees deployment validity. The broadcast degradation reported by BAMM [8] illustrates why a clean-file benchmark must state its domain rather than imply universal detection.

7.3 Limitations and Non-Claims

The primary cohort is modest, class-imbalanced, and concentrated in several generator families. Platform version labels are incomplete for supplementary tracks. Public lineage evidence cannot reconstruct every CLAM real-training identity, and content hashes cannot detect every perceptual derivative. The benchmark evaluates fully generated versus real music, not the degree or stage of AI assistance. Finally, the public research implementation of the Deezer method is not Deezer’s production detector [1].

8 Data and Code Availability

The version-pinned runner, public manifest, raw model scores, structured failure logs, analysis scripts, and checksums are released through GitHub and Hugging Face. The release is metadata-first and does not add a new audio bundle; each source must be acquired under its upstream license and verified against the published digest.

9 Ethics, Licensing, and Data Governance

Detector errors can harm creators if scores are treated as proof of authorship or fraud. ArtifactBench is intended to measure systems, not adjudicate individual works. Per-track public results use benchmark IDs rather than local paths or unnecessary creator-identifying strings. The release does not add a new bundle of platform- or YouTube-sourced audio; users retrieve source data under its own terms. FMA licenses remain attached per track, and non-commercial restrictions on SONICS, Deezer research code, and other upstream assets are preserved.

10 Conclusion

ArtifactBench v2 makes the denominator of AI-music detector evaluation explicit. By binding scores to content and lineage identity, freezing threshold selection before sealed testing, and reporting coverage separately from classification, it turns several hidden benchmark choices into inspectable artifacts. The result is not a universal test of synthetic music, but a reproducible way to locate where detectors generalize, where they fail, and which conclusions change when the cohort or operating point changes.

Relationship to Prior Work

ArtifactNet [9] introduced a residual-based detector and used an earlier ArtifactBench snapshot as one evaluation set. The present work treats the detector only as one baseline and contributes a lineage-aware benchmark design, audited release artifacts, and a comparative analysis of evaluation sensitivity.

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